

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (90 minutes) March 29th, 2014

Instructions: Attempt any three out of the following four questions on both pages.
Mention which three you would like to have graded. Open book and open notes.

1) The color of a person's eyes is determined by a single pair of genes. If they are both blue-eyed genes, then the person will have blue eyes; if they are both brown-eyed genes, then the person will have brown eyes; and if one of them is a blue-eyed gene and the other a brown-eyed gene, then the person will have brown eyes. A newborn child independently receives one gene from each of his parents and the gene he receives from a parent is equally likely to be either of the two eye genes of that parent. Suppose that Joe and both his parents have brown eyes, but Joe's sister and Joe's wife Joelle have blue eyes.

- What is the probability that Joe possesses a blue-eyed gene?
- What is the probability that Joe and Joelle's first child will have blue eyes?
- If Joe and Joelle's first child has brown eyes, what is the probability that their next child will also have brown eyes? *Hint: This is not the same as one minus the answer of part (b).*

2) a) Suppose that $X \sim \text{Laplace}(\sigma)$, with $\sigma > 0$; that is, the density of X is

$$f(x) = \frac{1}{2\sigma} e^{-|x|/\sigma}, \quad x \in \mathbb{R}.$$

What is the density of $|X|$?

b) Now suppose that $X \sim \text{exponential}(\lambda)$, with $\lambda > 0$; that is, the CDF of X is

$$F(x) = 1 - e^{-\lambda x}, \quad x > 0.$$

Let S be a random variable such that $P(S = -1) = P(S = 1) = 1/2$, and suppose that S is independent of X . Show that $SX \sim \text{Laplace}(1/\lambda)$.

3) Suppose that S is a $p \times p$ nonsingular sample covariance matrix which can be written as

$$S = \sum_{i=1}^p \lambda_i v_i v_i^T,$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ are called eigenvalues and $v_1, v_2, \dots, v_p$ are their corresponding eigenvectors. $v_1, v_2, \dots, v_p$ are orthogonal vectors of length 1.

Prove that the sequence $x_{n+1} = \sum x_n / |\sum x_n|$ converges to the eigenvector of the largest eigenvalue of S . Is there any condition on x_1 for this to be true?

4) The CDF of a chi-square distribution with 2 degrees of freedom is

$$F(y) = 1 - \exp(-y/2)$$

Given a Uniform (0,1) random variable U find the transformation $Y=h(U)$ such that Y has a chi-square distribution with 2 degrees of freedom.